

Recommendation **ITU-T K.52 (08/2024)**

SERIES K: Protection against interference

Guidance on complying with limits for human exposure to electromagnetic fields

Recommendation ITU-T K.52

Guidance on complying with limits for human exposure to electromagnetic fields

Summary

Recommendation ITU-T K.52 aims to help with compliance of telecommunication installations and mobile handsets, or other radiating devices used against the head with safety limits for human exposure to electromagnetic fields (EMFs). It presents general guidance, a calculation method and an installation assessment procedure. The assessment procedure for telecommunication installations, based on safety limits provided by the International Commission on Non-Ionizing Radiation Protection (ICNIRP), helps users determine the likelihood of installation compliance based on accessibility criteria, antenna properties and emitter power. The IEC Standard for the compliance measurement of mobile handsets is recommended.

History *

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Electronic attachment: Software calculator specified in Appendix III

Introduction

This Recommendation aims to help with compliance of telecommunication installations, such as base stations, and wireless communication devices, such as mobile handsets or other radiating devices used next to the ear or close to the body, with safety limits for human exposure to electromagnetic fields (EMFs). This Recommendation does not set safety limits; it seeks to provide techniques and procedures for assessing the compliance of telecommunication installations and handsets with national or international EMF safety limits.

Recommendation ITU-T K.52

Guidance on complying with limits for human exposure to electromagnetic fields

1 Scope

This Recommendation¹ aims to help with compliance of telecommunication installations, such as base station installations as defined in [IEC 62232], with safety limits for human exposure to electromagnetic fields (EMFs) produced by telecommunication equipment in the frequency range 8.3 kHz to 300 GHz². This Recommendation provides techniques and procedures for assessing the radiofrequency (RF) exposure to workers and the general public.

This Recommendation applies to hand-held and body-mounted wireless communication devices used next to the ear or close to the body operating in the frequency range 300 MHz to 6 GHz.

Where national laws, standards or guidelines on exposure limits to EMF exist and provide procedures that are at variance with this Recommendation, the pertinent national laws, standards or guidelines shall take precedence over the procedures provided in this Recommendation.

This Recommendation covers the exposure of people present on telecommunication sites, and the exposure of people outside telecommunication sites, to EMF produced by telecommunication equipment and equipment on telecommunication sites.

Contact current exposure, due to contact with conductive objects irradiated by electromagnetic fields, is not covered in this Recommendation.

Exposure due to the use of mobile handsets or other radiating devices, used in close proximity to the human body other than the head, is not covered.

[b-ITU-T Dir. VI] covers safety issues related to people getting in contact with telecommunication circuits exposed to the induction of a.c. electric power or a.c. electrified railway lines.

2 References

The following ITU-T Recommendations and other references contain provisions which, through reference in this text, constitute provisions of this Recommendation. At the time of publication, the editions indicated were valid. All Recommendations and other references are subject to revision; users of this Recommendation are therefore encouraged to investigate the possibility of applying the most recent edition of the Recommendations and other references listed below. A list of the currently valid ITU-T Recommendations is regularly published. The reference to a document within this Recommendation does not give it, as a stand-alone document, the status of a Recommendation.

[ITU-T K.61] Recommendation ITU-T K.61 (2018), *Guidance on Measurement and Numerical Prediction of Electromagnetic Fields for Compliance with Human Exposure Limits for Telecommunication Installations*.

[ITU-T K.100] Recommendation ITU-T K.100 (2024), *Measurement of radio frequency electromagnetic fields to determine compliance with human exposure limits when a base station is put into operation*.

¹ This Recommendation includes an electronic attachment with the software calculator specified in Appendix III.

² Appendix I also gives the exposure limits for lower frequencies.

- [IEC 62209-1528] IEC/IEEE 62209-1528:2020, *Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-worn wireless communication devices – Human models, instrumentation and procedures (Frequency range of 4 MHz to 10 GHz)*.
- [IEC 62209-3] IEC 62209-3:2019, *Measurement procedure for the assessment of specific absorption rate of human exposure to radio frequency fields from hand-held and body-mounted wireless communication devices – Part 3: Vector measurement-based systems (Frequency range of 600 MHz to 6 GHz)*.
- [IEC 62232] IEC 62232:2022, *Determination of RF field strength, power density and SAR in the vicinity of base stations for the purpose of evaluating human exposure*.
- [IEC 62479] IEC 62479:2010, *Assessment of the compliance of low-power electronic and electrical equipment with the basic restrictions related to human exposure to electromagnetic fields (10 MHz to 300 GHz)*.

3 Terms and definitions

3.1 Terms defined elsewhere

This Recommendation uses the following terms defined elsewhere:

3.1.1 actual value [IEC 62232]: Time-averaged value of transmitted power or equivalent isotropically radiated power (EIRP) of the equipment under test (EUT) using the averaging time specified in the applicable radio frequency (RF) exposure limits, e.g., 6 minutes using [b-ICNIRP].

3.1.2 actual maximum (max) value [IEC 62232]: Maximum value reached by the actual value when recorded by the EUT during a given period (e.g., week or month) (from [IEC 62232]).

3.1.3 actual maximum (max) approach [IEC 62232]: Evaluation method of RF exposure taking into account the actual maximum thereof.

3.1.4 actual maximum (max) threshold [IEC 62232]: Upper limiting value during base station (BS) operation of the actual transmitted power or EIRP configured by the EUT, for example using transmitted power or EIRP control features and tools implemented in the EUT.

3.1.5 average (temporal) power (P_{avg}) [IEC 62232]: Time-averaged rate of energy transfer defined by:

$$P_{avg} = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} P(t) dt$$

where t_1 and t_2 are the start and stop time of the exposure. The period $t_1 - t_2$ is the exposure duration time.

3.1.6 averaging time (T_{avg}) [IEC 62232]: Time duration specified in the applicable exposure limit guidelines over which exposure is averaged. Time duration specified in the applicable exposure limit guidelines over which exposure is averaged.

3.1.7 base station (BS) [IEC 62232]: Equipment, permanently or temporarily installed, including the radio transmitter(s) and associated antenna(s) as used in wireless telecommunications networks.

3.1.8 compliance boundary (CB) [IEC 62232]: Surface defining a volume outside which the radio frequency (RF) exposure from the equipment under test (EUT) is below the applicable exposure limit.

3.1.9 configured envelope radiation pattern [IEC 62232]: Non-physical radiation pattern obtained by taking, for each azimuth and elevation direction, the maximum performance among the set of the radiation patterns of the base station antenna configured during operation.

3.1.10 configured maximum (max) value [IEC 62232]: RMS maximum of the value (e.g., transmitted power or *EIRP*) as configured in the network manager of the base station.

3.1.11 directivity [IEC 62232]: Ratio of the radiation intensity produced by an antenna in a given direction to the value of the radiation intensities averaged across all directions in space.

3.1.12 equivalent isotropically radiated power (*EIRP*) [IEC 62232]: Product of the input power into an antenna and the antenna gain in a given direction relative to an isotropic antenna.

3.1.13 multiple-input, multiple-output (MIMO) [IEC 62232]: Technique used to increase the communication throughput by using transmission of two or more different data streams on two or more different antennas, using the same resources in both frequency and time, separated only by the use of different reference signals, and to be received by two or more antennas.

3.1.14 power reduction factor (F_{PR}) [IEC 62232]: Factor applied to the product of the configured maximum transmitted power or *EIRP* and the technology duty-cycle factor in order to obtain the actual maximum transmitted power or *EIRP*, or the actual maximum power or *EIRP* threshold, when a control feature is implemented.

NOTE – When applied to *EIRP*, the power reduction factor includes variations in both the transmitted power and the antenna gain.

3.1.15 technology duty-cycle factor (F_{TDC}) [IEC 62232]: Ratio of the time that a transmitter transmits over a specified period.

3.2 Terms defined in this Recommendation

This Recommendation defines the following terms:

3.2.1 antenna gain (G): Ratio of power radiated per unit solid angle multiplied by 4π to the total input power. Gain is frequently expressed in decibels with respect to an isotropic antenna (dBi). The equation defining gain is:

$$G(\theta, \varphi) = \frac{4\pi}{P_{in}} \frac{dP_r}{d\Omega}$$

where:

θ, φ the angles in a polar coordinate system

P_r the radiated power along the (θ, φ) direction

P_{in} the total input power

Ω elementary solid angle along the direction of observation.

3.2.2 continuous exposure: Exposure for a duration exceeding the corresponding averaging time. Exposure for less than the averaging time is called short-term exposure.

3.2.3 contact current: The current flowing into the human body when a person touches a conductive object in an electromagnetic field.

3.2.4 controlled/occupational exposure: Controlled/occupational exposure applies to situations where persons are exposed as a consequence of their employment and in which those persons who are exposed have been made fully aware of the potential for exposure and can exercise control over their exposure. Occupational/controlled exposure also applies where the exposure is of transient nature as a result of incidental passage through a location where the exposure limits may be above the general population/uncontrolled limits, as long as the exposed person has been made fully aware of the potential for exposure and can exercise control over his or her exposure by leaving the area or by some other appropriate means.

3.2.5 equipment under test (EUT): For the purpose of this Recommendation, the EUT is the base station that is put into operation, including all transmitting antennas (operating in the frequency range 100 MHz to 300 GHz).

3.2.6 exposure: Exposure occurs wherever a person is subjected to electric, magnetic or electromagnetic fields, or to contact currents other than those originating from physiological processes in the body or other natural phenomena.

3.2.7 exposure level: Exposure level is the value of the quantity used when a person is exposed to electromagnetic fields or contact currents.

3.2.8 exposure, non-uniform/partial body: Non-uniform or partial-body exposure levels result when fields are non-uniform over volumes comparable to the whole human body. This may occur due to highly directional sources, standing waves, scattered radiation or in the near field.

3.2.9 far-field region: That region of the field of an antenna where the angular field distribution is essentially independent of the distance from the antenna. In the far-field region, the field has a predominantly plane-wave character, i.e., locally uniform distribution of electric field strength and magnetic field strength in planes transverse to the direction of propagation.

3.2.10 general public: All non-workers (see definition of workers in clause 3.2.24) are defined as the general public.

3.2.11 induced current: Induced current is the current induced inside the body as a result of direct exposure to electric, magnetic or electromagnetic fields.

3.2.12 intentional emitter: A device that intentionally generates and emits electromagnetic energy by radiation or induction.

3.2.13 near-field region: The near-field region exists in proximity to an antenna or other radiating structure in which the electric and magnetic fields do not have a substantially plane-wave character but vary considerably from point-to-point. The near-field region is further subdivided into the reactive near-field region, which is closest to the radiating structure and that contains most or nearly all of the stored energy, and the radiating near-field region where the radiation field predominates over the reactive field, but lacks substantial plane-wave character and is complicated in structure.

NOTE – For many antennas, the outer boundary of the reactive near-field is taken to exist at a distance of one-half wavelength from the antenna surface.

3.2.14 power density (S): The power per unit area normal to the direction of electromagnetic wave propagation, usually expressed in units of Watts per square metre (W/m^2).

NOTE – For plane waves, power flux-density, electric field strength (E), and magnetic field strength (H) are related by the intrinsic impedance of free space, $\eta_0 = 377 \, \Omega$. In particular,

$$S = \frac{E^2}{\eta_0} = \eta_0 H^2 = EH$$

where E and H are expressed in units of V/m and A/m, respectively, and S in units of W/m^2 . Although many survey instruments indicate power density units, the actual quantities measured are E or H .

3.2.15 power density, average (temporal): The average power density is equal to the instantaneous power density integrated over a source repetition period.

NOTE – This averaging is not to be confused with the measurement averaging time.

3.2.16 power density, peak: The peak power density is the maximum instantaneous power density occurring when power is transmitted.

3.2.17 power density, plane-wave equivalent (S_{eq}): A commonly used term associated with any electromagnetic wave, equal in magnitude to the power flux-density of a plane wave having the same electric (E) or magnetic (H) field strength.

3.2.18 relative field pattern: The relative field pattern $f(\theta, \phi)$ is defined in this Recommendation as the ratio of the absolute value of the field strength (arbitrarily taken to be the electric field) to the absolute value of the maximum field strength. It is related to the relative numeric gain (see clause 3.2.19) as follows:

$$f(\theta, \phi) = \sqrt{F(\theta, \phi)}$$

3.2.19 relative numeric gain: The relative numeric gain $F(\theta, \phi)$ is the ratio of the antenna gain at each angle to the maximum antenna gain. It is a value ranging from 0 to 1. It is also called antenna pattern.

3.2.20 short-term exposure: Exposure for a duration less than the corresponding averaging time.

3.2.21 specific absorption (SA): Specific absorption is the quotient of the incremental energy (dW) absorbed by (dissipated in) an incremental mass (dm) contained in a volume element (dV) of a given density (ρ_m).

$$SA = \frac{dW}{dm} = \frac{1}{\rho_m} \frac{dW}{dV}$$

The specific absorption is expressed in units of joules per kilogram (J/kg).

3.2.22 specific absorption rate (SAR): The time derivative of the incremental energy (dW) absorbed by (dissipated in) an incremental mass (dm) contained in a volume element (dV) of a given mass density (ρ_m).

$$SAR = \frac{d}{dt} \frac{dW}{dm} = \frac{d}{dt} \frac{1}{\rho_m} \frac{dW}{dV}$$

SAR is expressed in units of watts per kilogram (W/kg).

SAR can be calculated by:

$$SAR = \frac{\sigma E^2}{\rho_m}$$

$$SAR = c \frac{dT}{dt}$$

$$SAR = \frac{j^2}{\rho_m \sigma}$$

where:

E the rms value of the electric field strength in body tissue in V/m

σ the conductivity of body tissue in S/m

ρ_m the density of body tissue in kg/m³

c the heat capacity of body tissue in J/kg°C

$\frac{dT}{dt}$ the time derivative of temperature in body tissue in °C/s

J the value of the induced current density in the body tissue in A/m².

3.2.23 general population/uncontrolled exposure: General population/uncontrolled exposure applies to situations in which the general public may be exposed, or in which persons who are exposed as a consequence of their employment may not be made fully aware of the potential for exposure, or cannot exercise control over their exposure.

3.2.24 workers: Employed and self-employed persons that are carrying out employment-related tasks.

3.2.25 unintentional emitter: A device that intentionally generates electromagnetic energy for use within the device, or that sends electromagnetic energy by conduction to other equipment, but which is not intended to emit or radiate electromagnetic energy by radiation or induction.

3.2.26 wavelength (λ): The wavelength of an electromagnetic wave is related to frequency (f) and velocity (v) of an electromagnetic wave by the following expression:

$$\lambda = \frac{v}{f}$$

In free space, the velocity is equal to the speed of light (c), which is approximately 3×10^8 m/s.

4 Abbreviations and acronyms

This Recommendation uses the following abbreviations and acronyms:

BS	Base Station
EIRP	Equivalent Isotropically Radiated Power
EMC	Electromagnetic Compatibility
EMF	Electromagnetic Field
ICNIRP	International Commission on Non-Ionizing Radiation Protection
LTE	Long Term Evolution
MIMO	Multiple-Input, Multiple-Output
NR	New Radio
SA	Specific Absorption
SAR	Specific Absorption Rate

5 Conventions

None.

6 General principles

There are many national and international documents that provide safety limits for human exposure to EMFs. Although these documents differ in particulars, most documents have several basic principles in common. These include the use of basic limits and reference levels, the use of two-tier exposure limits, averaging times, and separate consideration for exposure to low-frequency and high-frequency fields.

Most documents provide safety limits in terms of basic limits and reference (or derived) levels. The basic limits address the fundamental quantities that determine the physiological response of the human body to electromagnetic fields. Basic limits apply to a situation with the body present in the field. The basic limits for human exposure are expressed as the specific absorption rate (SAR), specific absorption (SA) and current density.

As the basic quantities are difficult to measure directly, most documents provide derived (reference) levels for electric field, magnetic field and power density. Reference levels may be exceeded if the exposure condition can be shown to produce SAR, SA, and induced current density below the basic limits. The reference levels apply to a situation where the electromagnetic field is not influenced by the presence of a body.

Most documents use a two-tier limit structure where lower levels are specified for uncontrolled/general public exposure than for controlled/occupational exposure.

It is important to emphasize that exposure limits are not emission limits; they apply to locations accessible to workers or members of the general public. Thus, it is possible to achieve compliance by limiting access to areas where the field limits are exceeded.

6.1 Multiple sources and frequencies

Most documents require that the effects of multiple sources be considered. Due to the different physiological effect of lower-frequency sources and higher-frequency sources, they should be considered separately. At frequencies typically below 10 MHz, the important physiological effects are due to the induced current density, while at frequencies typically above 100 kHz, the important physiological effects are due to the SAR.

To consider the effects of multiple sources, most documents require that the sources be considered in a weighted sum, where each individual source is pro-rated according to the limit applicable to its frequency. Appendix I shows the procedure in the International Commission on Non-Ionizing Radiation Protection (ICNIRP) guidelines.

6.2 Exposure duration

Most documents define the exposure limits in terms of quantities averaged over a time period called the averaging time. In case of short-term exposure with duration less than the averaging time, the applicable limit is:

$$\sum_i X_i^2 t_i \leq X_l^2 t_{avg}$$

where:

X_i the field (E or H) during exposure i

t_i the duration of exposure i

X_l the reference limit

t_{avg} the appropriate averaging time.

The power density limit is:

$$\sum_i S_i t_i \leq S_l t_{avg}$$

where:

S_i the power density during exposure i

t_i the duration of exposure i

S_l the reference limit

t_{avg} the appropriate averaging time.

7 EMF safety limits

In many cases, local or national regulatory agencies or standards bodies promulgate the EMF safety limits. If such limits do not exist, or if they do not cover the frequencies of interest, it is recommended that the limits from [b-ICNIRP] (Appendix I) be used.

8 Compliance with wireless communication devices

For wireless communication devices operating in the frequency range of 300 MHz to 10 GHz, compliance with the exposure limits from [b-ICNIRP] can be achieved by applying the measurement procedures for SAR in [IEC 62209-1528] and [IEC 62209-3]. Also, in certain cases, local or national regulatory agencies or standards bodies may recommend national or regional measurement practices in the spirit of [IEC 62209-1528] to get a SAR value for wireless communication devices used next to the ear.

9 Achieving compliance to EMF safety limits for telecommunication installations

The following steps should be taken to achieve compliance:

- 1) Identify appropriate compliance limits.
- 2) Determine if EMF exposure assessment for the installation of equipment in question is needed. (See clause 9.1.)
- 3) If the EMF exposure assessment is needed, it may be performed by calculations or measurement. This Recommendation presents a risk assessment approach to help the user determine the possibility that EMF exposure limits may be exceeded and help the user select an appropriate method to perform the assessment.
- 4) If the EMF exposure assessment indicates that pertinent exposure limits may be exceeded in areas where people may be present, mitigation/avoidance measures should be applied.

9.1 Determining the need for assessment for telecommunication equipment

Telecommunication equipment should be classified as an intentional or unintentional EMF emitter in accordance with the definitions. Typically, an intentional emitter is associated with an antenna for radiation of electromagnetic energy.

9.1.1 Unintentional emitters

Unintentional emitters may produce EMF due to spurious emissions. There are electromagnetic compatibility (EMC) emission standards that limit the magnitude of these spurious fields. Typically, the fields produced by telecommunication equipment that is an unintentional emitter are significantly below the safety limits established by ICNIRP and national standards. The limits established for EMC compliance are orders of magnitude below the EMF safety limits. Even if equipment exceeds the emission limits at certain frequencies, experience indicates that the fields produced are still orders of magnitude below the safety limits. Thus, telecommunication equipment that is an unintentional emitter does not need an EMF safety assessment to assure compliance with safety limits.

9.1.2 Intentional emitters

Intentional emitters use electromagnetic fields for signal transmission. They produce EMF that may exceed the safety limits in some regions depending on the operating power, gain, frequency, orientation and directivity of the transmitting antenna. It is possible to take into account these parameters, and the operating environment of the installation, to determine the need and the appropriate procedure of exposure assessment. This Recommendation presents a risk assessment approach based on classification of exposure zones. As an example, a base station is an intentional emitter.

9.2 Procedures for EMF exposure assessment

If it is determined that an EMF exposure assessment is needed due to the presence of intentional emitters, it should be performed for all locations where people might be exposed to EMF. The intent of the assessment is to classify potential exposure to EMF as belonging to one of the three following zones:

- 1) **Compliance zone:** In the compliance zone, potential exposure to EMF is below the applicable limits for both controlled/occupational exposure and uncontrolled/general public exposure.
- 2) **Occupational zone:** In the occupational zone, potential exposure to EMF is below the applicable limits for controlled/occupational exposure but exceeds the applicable limits for uncontrolled/general public exposure.
- 3) **Exceedance zone:** In the exceedance zone, potential exposure to EMF exceeds the applicable limits for both controlled/occupational exposure and uncontrolled/general public exposure.

For many installations, the exceedance zone and the occupational zone are not accessible to people, or are only accessible under unusual circumstances, such as a person standing directly in front of the antenna. The risk assessment procedure presented in this Recommendation is concerned primarily with exposure of the general public, and workers in the course of their normal activities. See Figure 1.

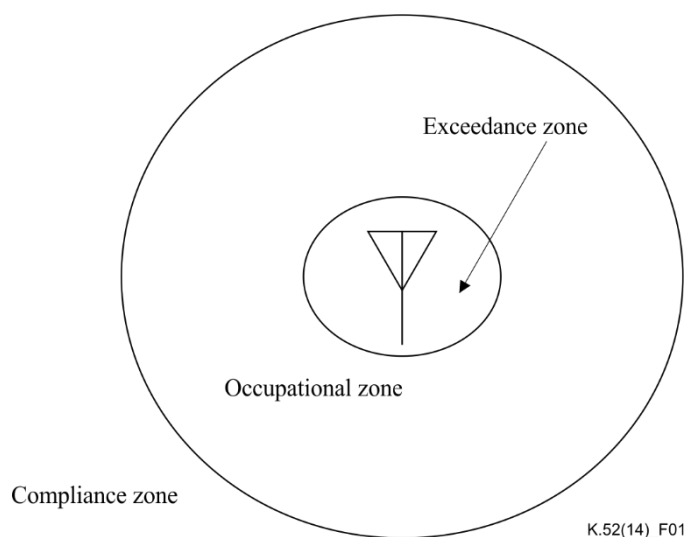


Figure 1 – Figurative illustration of exposure zones

The determination of the exposure zones should follow the methods defined in [IEC 62232].

9.3 RF EMF exposure assessment procedure for base stations

9.3.1 General guidelines

[IEC 62232] defines methods for RF exposure evaluation for three main applications:

- Base station (BS) compliance: determination of compliance boundary information for BS before it is placed on the market.
- BS installation compliance: determination of the total RF exposure levels in accessible areas from a BS and other relevant sources before the EUT is put into operation.
- In-situ RF exposure assessment: measurement of RF exposure levels in the vicinity of a BS installation after the EUT has been installed and is operating

The recommendations in this clause are related to BS installation type application.

The assessment of the exposure level shall consider:

- either the configured maximum or the actual maximum transmitted power or equivalent isotropically radiated power (EIRP) of the BS, see definitions in clause 3 and [IEC 62232];
- the simultaneous presence of other EMF sources, even at different frequencies.

The following information should also be considered:

- BS technology (e.g., long term evolution (LTE), new radio (NR)) and system parameters (e.g., technology duty cycle factor, massive MIMO, etc.);
- the frequency of operation;
- the antenna gain G (see definition in clause 3.2.1) or the relative numeric gain F (see definition in clause 3.2.19), including maximum gain and beam width; and
- various characteristics of the installation, such as the antenna location, antenna height, beam direction, beam tilt and access restrictions.

The actual maximum transmitted power or EIRP is derived from the configured maximum power or EIRP using the following equations:

$$P_{\text{TXAM}} = P_{\text{TXM}} \cdot F_{\text{TDC}} \cdot F_{\text{PR}} \quad (1)$$

$$EIRP_{\text{TXAM}} = P_{\text{TXM}} \cdot F_{\text{TDC}} \cdot G_{\text{MLB}} \cdot F_{\text{PR}} \quad (2)$$

where:

- P_{TXAM} is the actual maximum transmitted power;
- P_{TXM} is the maximum transmitted power configured in downlink (deterministic factor);
- $EIRP_{\text{TXAM}}$ is the actual maximum transmitted EIRP (per sector, cell or segment);
- G_{MLB} is the maximum gain in the main lobe for fixed beam or boresight for steerable beams (deterministic factor);
- F_{TDC} is the technology duty cycle factor;
- F_{PR} is the power reduction factor.

9.3.2 The installation category scheme

Each emitter installation should be classified into one of the following three categories:

- 1) **Inherently compliant:** Inherently safe sources produce fields that comply with the applicable exposure limits a few centimetres away from the source, see for example [IEC 62479] or E0 and E2 installation classes described in [IEC 62232]. Particular precautions are not necessary.
- 2) **Normally compliant:** Normally compliant installations contain sources that produce EMF that can exceed the applicable exposure limits. However, as a result of normal installation practices and the typical use of these sources for communication purposes, the exceedance zone of these sources is not accessible to the general public under ordinary conditions, see for example E10, E100 and E+ installation classes described in [IEC 62232]. Additional examples include antennas mounted on sufficiently tall towers or narrow-beam earth stations pointed at the satellite. Precaution may need to be exercised by maintenance personnel who come into the close vicinity of emitters in certain normally compliant installations.
- 3) **Provisionally compliant:** These installations require special measures to achieve compliance. This involves determination of the exposure zones and measures presented in clause 10.

9.3.3 Procedure for determining the installation category

Each installation should be categorized into one of the installation classes described in [IEC 62232] or [ITU K.100] or installation categories as defined in clause 9.3.2.

It is expected that operators providing a particular telecommunication service use a limited set of antennas and associated equipment with well-defined characteristics. Furthermore, installation and exposure conditions for many emitter sites are likely to be similar. Therefore, it is possible to define

a set of reference configurations, conditions and corresponding critical parameters that will enable convenient classification of sites.

A useful procedure is as follows:

- 1) Define a set of reference antenna parameters or antenna types. These types can be customized to the types of emitters used for the particular application.
- 2) Define a set of accessibility conditions. These categories depend on the accessibility of various areas in the proximity of the emitter to people. These categories can be customized to the most commonly occurring installation environment for the particular service or application.
- 3) For each combination of reference antenna parameters and accessibility condition, determine the threshold EIRP. This threshold EIRP, which will be denoted as $EIRP_{th}$, is the value that corresponds to the exposure limit for the power density or field from the reference antenna for the accessibility condition. The determination may be performed by calculation or measurements as described in clause 9.3.4. Provided the categories are sufficiently encompassing, this determination need only be performed once for the majority of installations.
- 4) An installation source belongs to the inherently compliant class if the emitter is inherently compliant (as defined above). There is no need to consider other installation aspects.
NOTE – Appendix IV shows that a source inherently compliant with [b-ICNIRP] limits has EIRP less than 2 W.
- 5) For each site, an installation belongs to the normally compliant class, if the following criterion is fulfilled:

$$\sum_i \frac{EIRP_i}{EIRP_{th,i}} \leq 1$$

where $EIRP_i$ is the temporal averaged radiated power of the antenna at a particular frequency i , and $EIRP_{th,i}$ is the EIRP threshold relevant to the particular antenna parameters and accessibility conditions. For a multiple-antenna installation, the following two conditions need to be distinguished:

- If the sources have overlapping radiation patterns as determined by considering the half-power beam width, the respective maximum time-averaged EIRP should satisfy the criterion.
 - If there is no overlap of the multiple sources, they shall be considered independently.
- 6) Sites that do not meet the conditions for normally compliant classification are considered provisionally compliant.

For sites where the application of these categories is ambiguous, additional calculations or measurements will need to be performed.

Annex B presents a set of basic configurations, exposure conditions, parameters and relevant $EIRP_{th}$ values. The set of Annex B should be used as a default unless the operator defines another set that is appropriate for a given service deployment and performs the relevant exposure analysis. Appendix III provides examples of $EIRP_{th}$ calculation.

9.3.4 Determination of the $EIRP_{th}$

The procedure for determining $EIRP_{th}$ is the following:

- 1) Determine the field or the power density for each point O, where exposure can occur, for the particular antenna.
- 2) Find the maximum power density S_{max} within the exposure area from this set.

- 3) The condition $S_{max} = S_{lim}$ gives the $EIRP_{th}$ where S_{lim} is the relevant limit given by the EMF exposure standard at the relevant frequency.

This procedure may be performed by the calculations shown in clause 10.1, by other more accurate calculation methods or by measurements. If measurements are used, it is necessary to perform them at a number of representative locations for each accessibility configuration and antenna type.

For BS installations with massive MIMO antennas, the basic criteria described in Annex B and the $EIRP_{th}$ calculation examples described in Appendix III are applicable provided that the antenna radiation pattern is represented by the configured envelop radiation pattern of the available traffic beams, see also [IEC 62232]. The half-power beamwidth, side-lobe level and tilt parameters should be selected to be representative of the configured envelope radiation pattern. To determine $EIRP_{th}$ values relevant for actual maximum exposure, a power reduction factor (F_{PR}) should be applied as described in 8.3.1 and [IEC 62232].

9.4 Uncertainty

Uncertainty analysis shall be performed in accordance with clause 10.3. Additional information is provided in [ITU-T K.100] and [IEC 62232]

For base stations, the target expanded uncertainty is less than or equal to 4 dB, which is considered industry best practice. The expanded uncertainty for the RF exposure evaluation used for the product installation compliance assessment shall not exceed 6 dB. Under this condition, the evaluation result shall be compared directly to the exposure limit. However, the entity performing the uncertainty analysis also needs to be aware of the applicable regulation.

10 EMF evaluation techniques

This clause presents methods that can be used to evaluate EMF for telecommunication installations. Additional information for terrestrial broadcasting systems may be found in [b-ITU-R. BS.1698].

10.1 Calculation methods

In addition to the basic analytical methods described in this clause, [IEC 62232] and [ITU-T K.61] provide guidance on the selection of numerical methods suitable for EMF exposure prediction in various situations.

10.1.1 Reactive near-field region

In the reactive near-field region, the electric and magnetic fields must be considered separately. In the absence of field-distorting objects, the fields can be calculated using quasi-static formulae if a current distribution is known.

10.1.2 Far-field region

The following material provides methods for conservatively estimating field strength and power density levels.

For a single radiating antenna, the approximate power density radiated in the direction described by the angles θ (complementary to the elevation angle) and ϕ (azimuth angle) can be evaluated by the following expression:

$$S(R, \theta, \phi) = \frac{EIRP}{4\pi} \left[f(\theta, \phi) \frac{1}{R} + \rho f(\theta', \phi') \frac{1}{R'} \right]^2$$

where:

$S(R, \theta, \phi)$ the power density in W/m²

$f(\theta, \phi)$ the relative field pattern of the antenna (positive number between 0 and 1)

EIRP the *EIRP* of the antenna in W

ρ the absolute value (modulus) of the reflection coefficient and takes into account the wave reflected by the ground. In some cases, the exposure to the reflected wave may be blocked, so that ρ should be set to 0

R the distance between the central point of the radiating source and the putative exposed person

R' the distance between the central point of the image of the radiating source and the putative exposed person.

Near ground level, the values of primed variables are approximately equal to the unprimed, so the power can be calculated as:

$$S_{gl}(R, \theta, \phi) = (1 + \rho)^2 \frac{EIRP}{4\pi R^2} F(\theta, \phi)$$

where:

$F(\theta, \phi)$ the relative numeric gain of the antenna relative to an isotropic radiator (positive number between 0 and 1).

The reflection coefficient ρ of earth with conductivity σ , permittivity $\varepsilon = \kappa \varepsilon_0$ (ε_0 = permittivity of vacuum, κ = relative permittivity) and grazing angle of incidence Ψ is:

$$\rho = \frac{(\kappa - j\chi) \sin \psi - \sqrt{(\kappa - j\chi) - \cos^2 \psi}}{(\kappa - j\chi) \sin \psi + \sqrt{(\kappa - j\chi) - \cos^2 \psi}} \quad \text{vertical polarization}$$

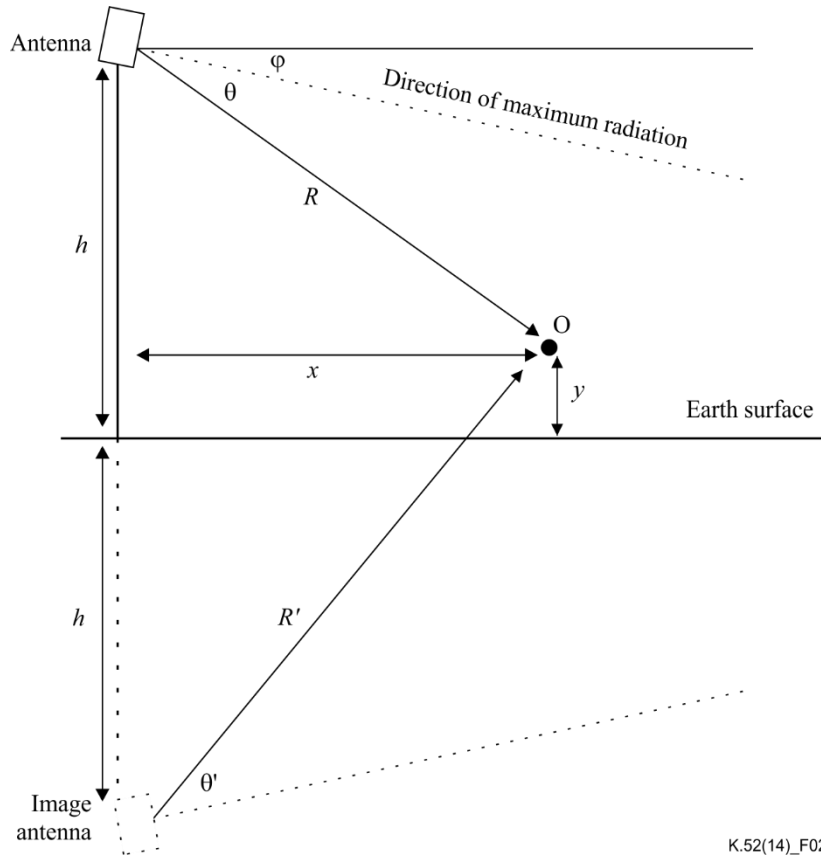
$$\rho = \frac{\sin \psi - \sqrt{(\kappa - j\chi) - \cos^2 \psi}}{\sin \psi + \sqrt{(\kappa - j\chi) - \cos^2 \psi}} \quad \text{horizontal polarization}$$

where:

$$\chi = \frac{\sigma}{\omega \varepsilon_0}$$

In general, the reflected wave contains components in vertical and horizontal polarization that vary with the incidence angle. However, for many applications, it is sufficient to consider only the predominant polarization of the incident wave in calculating the reflection coefficient.

The distances and angles are defined in Figure 2. It is assumed that exposure is being evaluated at point O.



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Figure 2 – Definition of distances and vertical angles

For rooftop locations, attenuation caused by building materials in the walls and roof can reduce the exposure inside a building by at least 10-20 dB.

The electric and magnetic fields are calculated using:

$$E = \sqrt{S\eta_0}$$

$$H = \sqrt{S/\eta_0}$$

where $\eta_0 = 377 \, \Omega$ is the intrinsic impedance of free space.

The foregoing equations are valid for the far-field region. Their use in the near-field region may yield inaccurate (overly conservative) results. Thus, these equations can be used to determine compliance with the EMF exposure limits.

10.2 Measurement procedures

Measurements are useful in cases where the fields are difficult to calculate and where the calculations yield values that are near the exposure limit threshold. [IEC 62232] and [ITU-T K.61] provide guidance on measurement methods that can be used to verify compliance with EMF exposure standards. In addition, any applicable national standards should be consulted for guidance about EMF measurements. A number of other publications also listed in the Bibliography provide detailed information about measuring EMF fields at various frequencies. For example, [b-IEC 61786] can be consulted for measurements below 100 kHz.

10.3 Uncertainty

Uncertainty shall be estimated for every reported measured and calculated RF field strength, power density or SAR evaluation. Guidance on how to estimate uncertainty can be found in

[IEC 62209-1528], [IEC 62209-3], [IEC 62232] and [b-JCGM 100]. Although [b-JCGM 100] is concerned with measurement uncertainty, the same concepts can be extended to computational uncertainty.

11 Mitigation techniques

It is necessary to control EMF exposure in locations accessible to people where the EMF exposure exceeds the applicable EMF exposure limits. An effective way to control exposure where other installations characteristics cannot be changed, is to restrict access to areas where the limits are exceeded.

11.1 Occupational zone

If the EMF exposure exceeds the applicable limits for uncontrolled/general public exposure but does not exceed the limits for occupational exposure, then access to the general public should be restricted, but workers may be permitted to enter the area. Physical barriers, lockout procedures or adequate signs can accomplish the access restriction. Workers entering the occupational zone should be informed.

It is recommended not to locate a permanent workplace within the occupational zone.

11.2 Exceedance zone

Where the EMF exposure exceeds the applicable limits for occupational exposure, access to workers and the general public should be restricted. If it is necessary for workers to enter the area, steps to control their exposure should be taken. Such steps include:

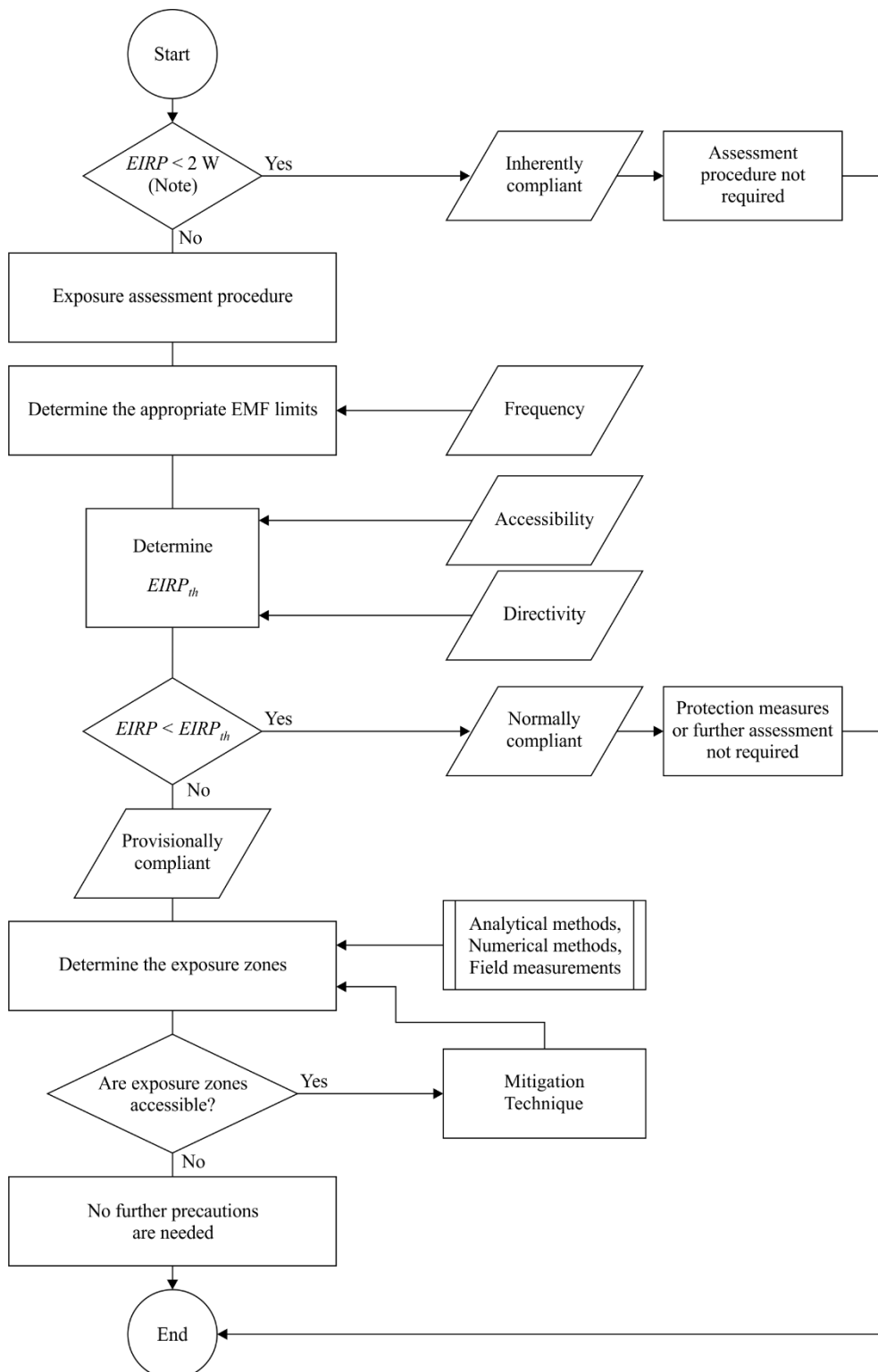
- temporarily reducing the power of the emitter;
- controlling the duration of the exposure so that time-averaged exposure is within safety limits;
- shielding or use of protective clothing.

Annex A

The application flow chart

(This annex forms an integral part of this Recommendation.)

This annex shows an example of a flow chart of the exposure assessment for a single EMF source of a telecommunication installation. The flow chart is intended for telecommunication infrastructure equipment, such as base station or earth station, only. The BS installation evaluation process flow chart provided in [IEC 62232] also applies.



NOTE – See Appendix IV.

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Annex B

Basic criteria for determining the installation category

(This annex forms an integral part of this Recommendation.)

Installation classes have been defined in [IEC 62232] and [ITU-T K.100] to provide simplified RF EMF exposure assessment of BS installations. For example, the installation class E0 corresponds to product that comply with [IEC 62479] or products that have zero compliance boundary dimensions.

This annex provides additional criteria for the classification of the installation on the basis of the [b-ICNIRP] limits. The criteria are based on a conservative estimate of the likely EMF exposure in the various situations described below.

B.1 Inherently compliant sources

Base stations emitters with a maximum EIRP of 2 W or less are classified as inherently compliant. No further action is deemed necessary.

If the total radiating power is 100 mW or less and the antenna(s) are low-gain small-aperture microwave or millimetre-wave antennas, the emitter can be regarded as inherently compliant. No further action is deemed necessary.

In addition, where the emitter is so constructed that access to any area where exposure limits may be exceeded is precluded by the construction of the radiating device, is classified as inherently compliant.

B.2 Normally compliant installations

The suggested criteria for determining if an installation is normally compliant comprises three installation characteristics: the accessibility and the directivity of the antenna, the frequency of the radiated field. These characteristics are described in clauses B.2.1, B.2.2 and B.2.3.

The $EIRP_{th}$ values to be compared by the $EIRP$ of the installation can be determined by considering the above characteristics. A possible way to define the $EIRP_{th}$ is described in Appendix III.

B.2.1 Accessibility categories

This clause defines the accessibility categories. These categories, which depend on the installation circumstances, assess the likelihood that a person can access the exceedance zone of the emitter. See Table B.1.

Table B.1 – Accessibility categories

Accessibility category	Relevant installation circumstances	Figure reference
1	Antenna is installed on an inaccessible tower – the centre of radiation is at a height h above ground level. There is a constraint $h > 3$ m. Antenna is installed on a publicly accessible structure (such as a rooftop) – the centre of radiation is at a height h above the structure.	Figure B.1

Table B.1 – Accessibility categories

Accessibility category	Relevant installation circumstances	Figure reference
2	Antenna is installed at ground level – the centre of radiation is at a height h above ground level. There is an adjacent building or structure accessible to the general public and of approximately height h located a distance d from the antenna along the direction of propagation. There is a constraint $h > 3$ m.	Figure B.2
3	Antenna is installed at ground level – the centre of radiation is at a height h ($h > 3$ m) above ground level. There is an adjacent building or structure accessible to the general public and of approximately height h' located at a distance d from the antenna along the direction of propagation.	Figure B.3
4	Antenna is installed on a structure at a height h ($h > 3$ m). There is an exclusion area associated with the antenna. Two geometries for the exclusion area are defined: – A circular area with radius a surrounding the antenna; or – A rectangular area of size $a \times b$ in front of the antenna.	Figure B.4 Figure B.5

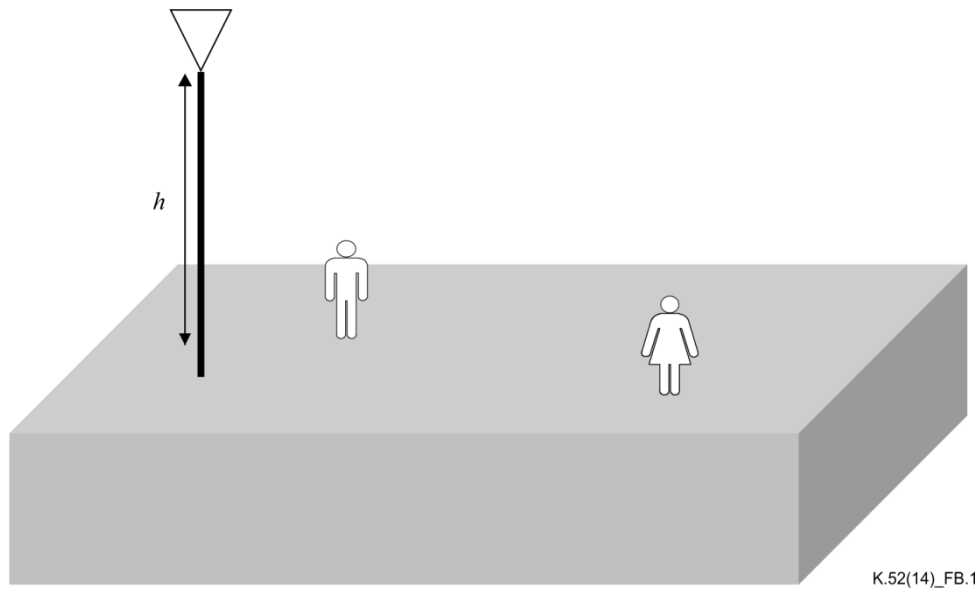
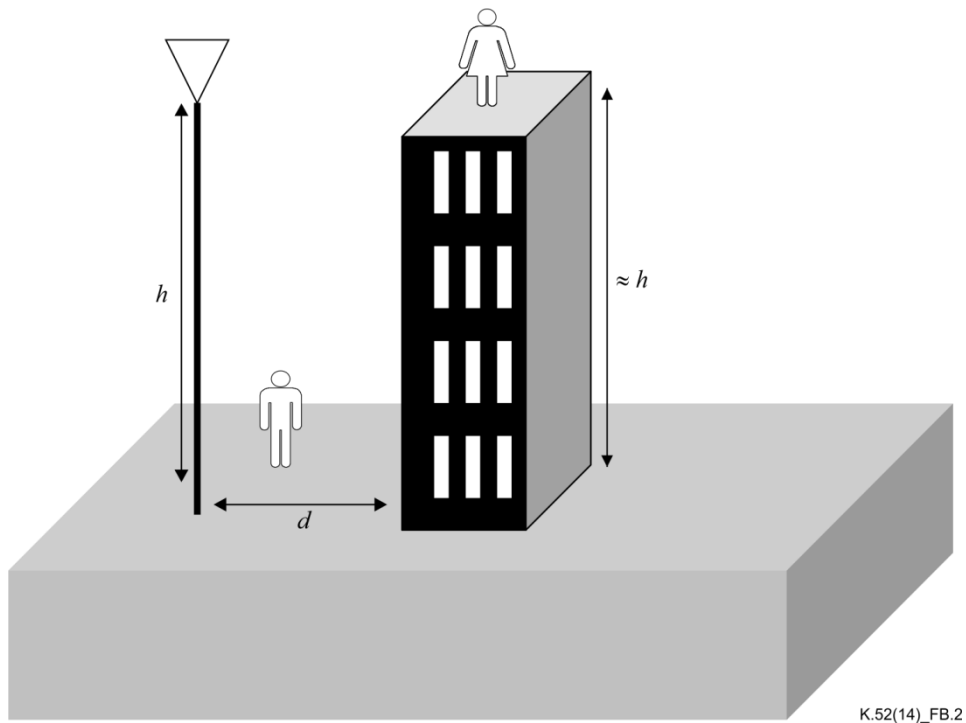
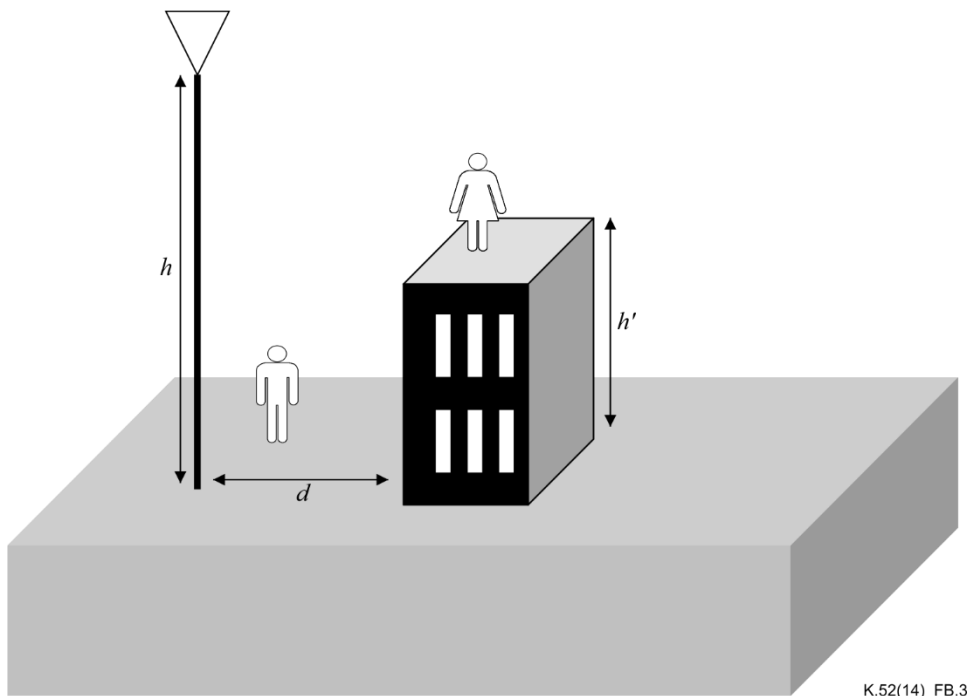


Figure B.1 – Illustration of the accessibility category 1



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Figure B.2 – Illustration of the accessibility category 2



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Figure B.3 – Illustration of the accessibility category 3

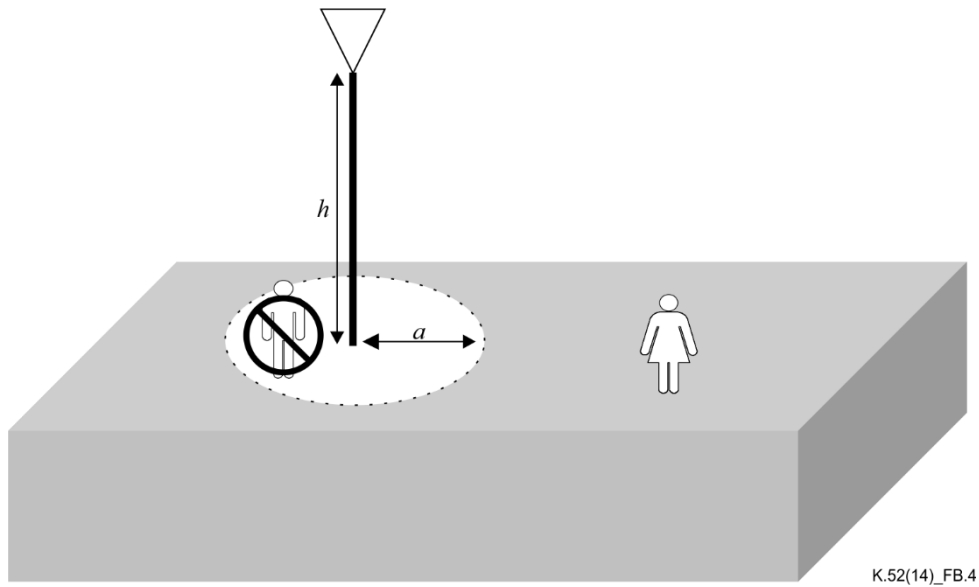


Figure B.4 – Illustration of the accessibility category 4, circular exclusion area

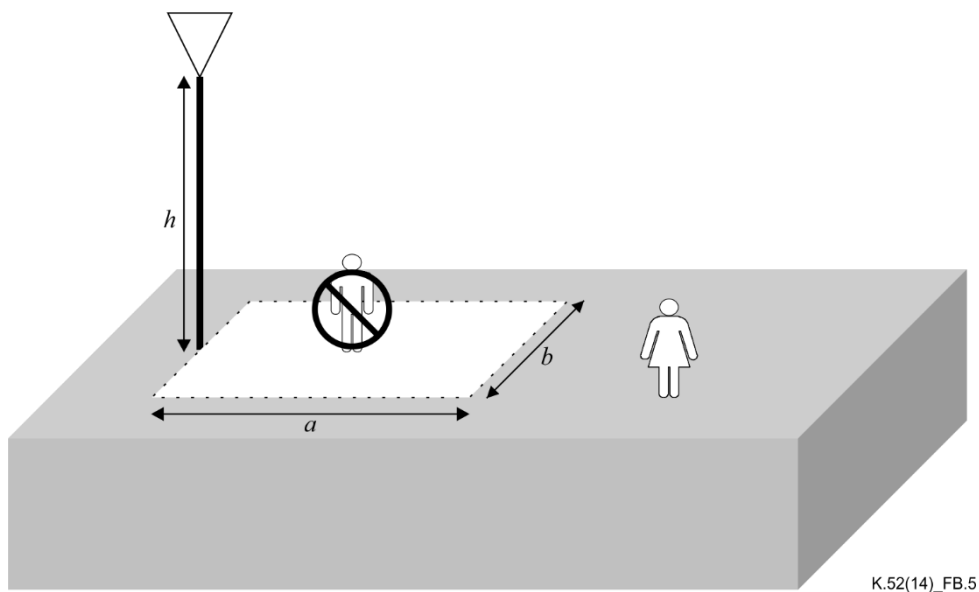


Figure B.5 – Illustration of the accessibility category 4, rectangular exclusion area

B.2.2 Frequency ranges

The carrier frequency determines the exposure limit for the radiated power density, $S_{lim}(f)$ as reported in the electromagnetic fields exposure standards.

B.2.3 Antenna directivity categories

Antenna directivity is important because it determines the pattern of potential exposure. High directivity means that most of the radiated power is concentrated in a narrow beam which may allow good control of the location of the exposure zones.

The antenna pattern is a major determinant and a frequently varying factor in determining the field. Table B.2 presents a description to facilitate classification of antennas into generic categories. The most important parameter for determining the exposure due to elevated antennas is the vertical (elevation) antenna pattern. The horizontal (azimuth) pattern is not relevant because the exposure assessment assumes exposure along the direction of maximum radiation in the horizontal plane.

Note, however, that the vertical and horizontal patterns determine the antenna gain, and that horizontal pattern determines the exclusion area for accessibility category 4.

Table B.2 – Antenna directivity categories

Directivity category	Antenna description	Relevant parameters
1	Half-wave dipole	None See Figure B.6
2	Broad coverage antenna (omnidirectional or sectional), such as those used for wireless communication or broadcasting	- Vertical half-power beamwidth: θ_{bw} - Maximum side-lobe amplitude with respect to the maximum: A_{sl} - Beam tilt: α See Figure B.7
3	High-gain antenna producing a "pencil" (circularly symmetrical beam), such as those used for point-to-point communication or earth stations	- Vertical half-power beamwidth: θ_{bw} - Maximum side-lobe amplitude with respect to the maximum: A_{sl} - Beam tilt: α See Figure B.7
NOTE – A_{sl} should be expressed as a numerical factor. However, usually, it is given in dB with respect to the maximum. To convert: $A_{sl} = 10^{A_{sl}[dB]/10}$.		

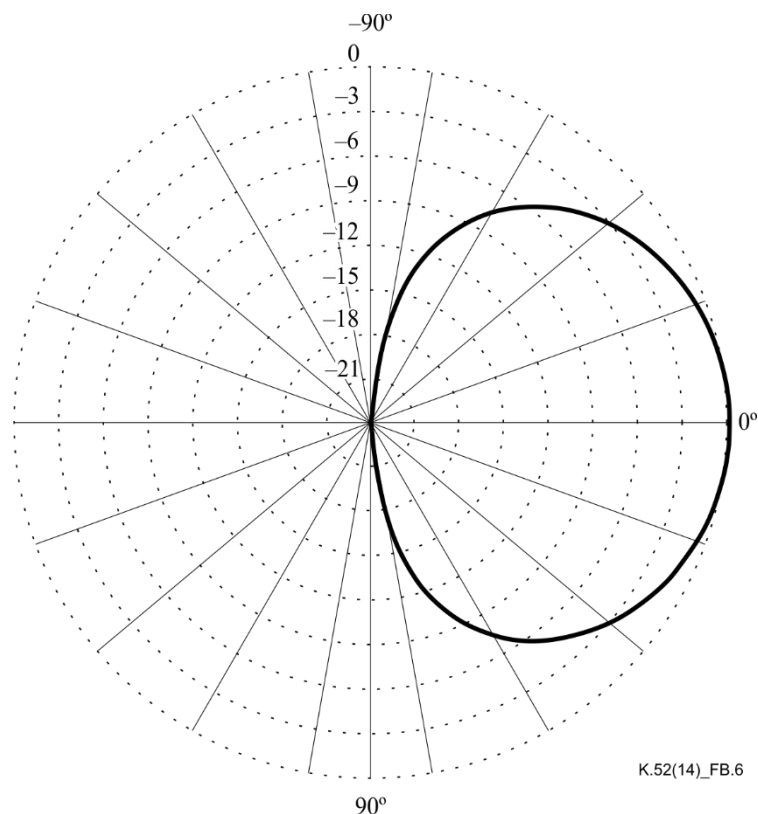


Figure B.6 – Vertical pattern for a half-wave dipole in vertical polarization

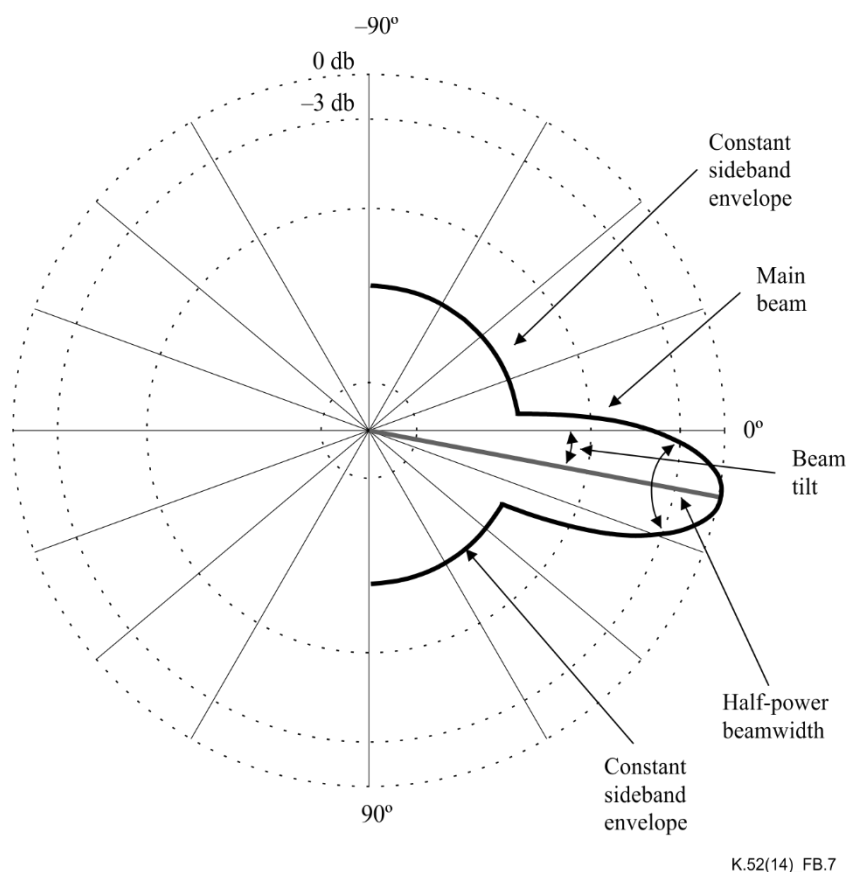


Figure B.7 – Illustration of terms relating to antenna patterns

B.2.4 The exclusion area

This clause describes the exclusion areas for accessibility category 4. The exclusion area depends on the horizontal pattern of the antenna. The relevant parameter is the horizontal coverage of the antenna. Table B.3 presents the exclusion areas for a few typical values of the horizontal coverage of omnidirectional, sectional or narrow-beam antennas.

Table B.3 – Exclusion area as function of horizontal coverage

Horizontal coverage	Exclusion area
Omnidirectional	Circular area (Figure B.4)
120°	Rectangular area (Figure B.5) $b = 0.866a$
90°	Rectangular area (Figure B.5) $b = 0.707a$
60°	Rectangular area (Figure B.5) $b = 0.5a$
30°	Rectangular area (Figure B.5) $b = 0.259a$
Less than 5°	Rectangular area (Figure B.5) $b = 0.09a$

Appendix I

RF EMF exposure limits

(This appendix does not form an integral part of this Recommendation.)

Administrations are encouraged to follow the ICNIRP Guidelines [b-ICNIRP] or IEEE Standard [b-IEEE C95.1], or science-based RF EMF exposure limits set by their own experts. The best practice for Administrations that choose to use international RF EMF exposure limits is to limit the exposure levels to the thresholds specified in [b-ICNIRP]. [b-ICNIRP] and [b-IEEE C95.1] limits are largely harmonized, and the power-density limits for whole-body exposure to continuous fields are identical above 30 MHz. See [b-ITU-T K.91].

Appendix II

Example of simple evaluation of EMF exposure

(This appendix does not form an integral part of this Recommendation.)

This appendix presents an example of using a simple prediction method to evaluate EMF exposure.

II.1 Exposure at the ground level

The geometry for calculating exposure at the ground level due to an elevated antenna is shown in Figure II.1.

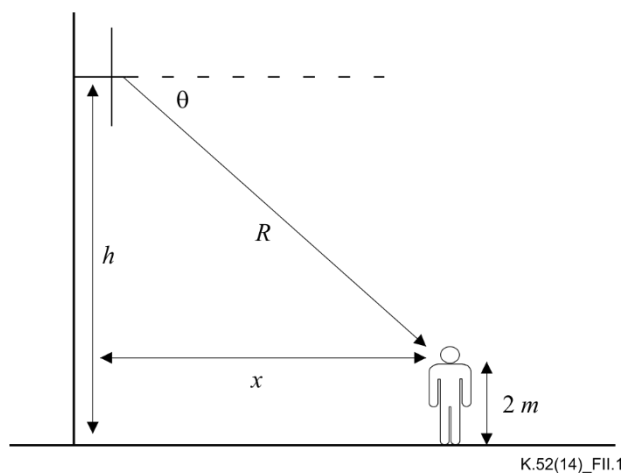


Figure II.1 – Sample configuration for calculating exposure at ground level

An antenna is installed so that the centre of radiation is at the height h above the ground. The goal of the calculation is to evaluate the power density at a point 2 m above the ground (approximate head level) at a distance x from the tower. In this example the main beam is parallel to the ground and the antenna gain is axially symmetrical (omnidirectional).

To simplify the foregoing, define $h' = h - 2$ [m]. Using trigonometry:

$$R^2 = h'^2 + x^2$$

$$\theta = \tan^{-1}\left(\frac{h'}{x}\right)$$

Taking into account reflections from the ground, the power density becomes:

$$S = \frac{2.56}{4\pi} F(\theta) \frac{EIRP}{x^2 + h'^2}$$

NOTE – The factor of 2.56 could be replaced by 4 (i.e., considering a reflection factor of 1) if a more severe approach is necessary.

For example, if the antenna is a half-wave dipole, the relative numeric gain is of the form of:

$$F(\theta, \phi) = \left[\frac{\cos\left(\frac{\pi}{2} \sin \theta\right)}{\cos \theta} \right]^2$$

Then, for a source with $EIRP$ of 1 000 W, the exposure power as a function of x is shown in Figure II.2 for three different heights.

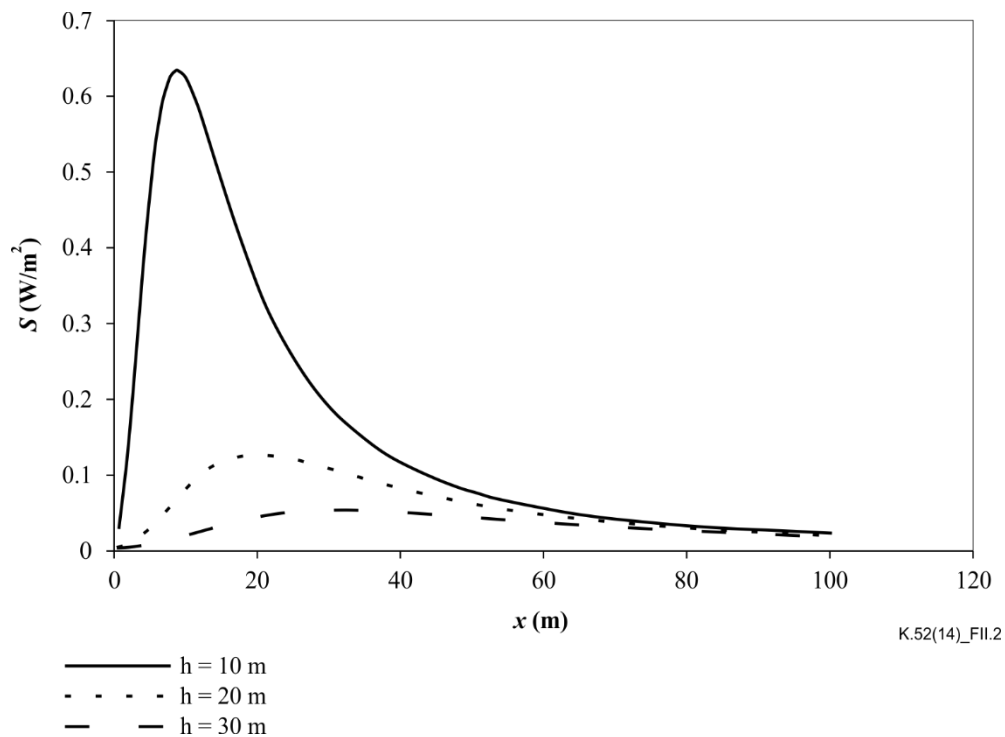


Figure II.2 – Power density at the ground level vs distance from the tower calculated for the example in Figure II.1

II.2 Exposure at an adjacent building

The geometry for calculating exposure at a building adjacent to an antenna tower is shown in Figure II.3.

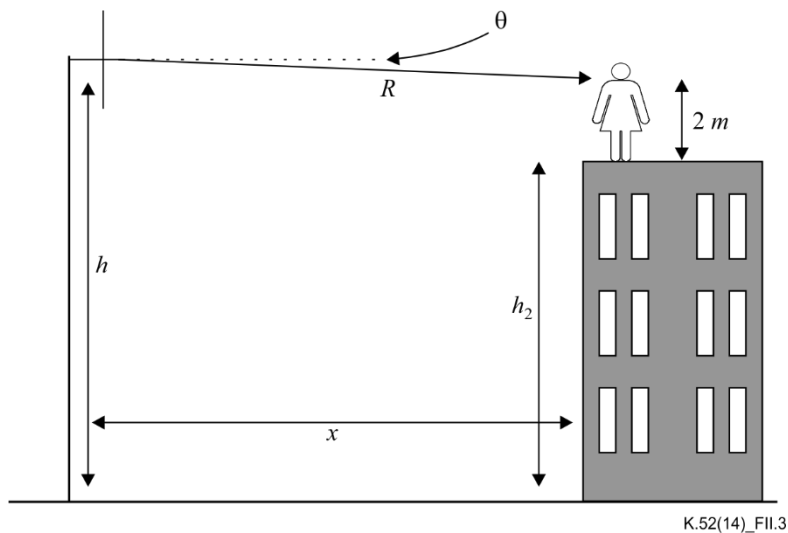


Figure II.3 – Sample configuration for calculating exposure at an adjacent building

An antenna is installed so the centre of radiation is at the height h above the ground. The goal of the calculation is to evaluate the power density at a point 2 m above the roof level (approximate head level) of an adjacent building. The building has a height h_2 and is located at a distance x from the tower. The most severe exposure is expected at the edge of the roof closest to the antenna. It is assumed that the main beam is parallel to the ground and that the antenna gain is axially symmetrical (omnidirectional).

Again, to simplify the foregoing, define $h' = h - h_2 - 2$. Using trigonometry:

$$R^2 = h'^2 + x^2$$

$$\theta = \tan^{-1}\left(\frac{h'}{x}\right)$$

In this situation, the reflections from the ground may be neglected since the reflected wave is likely to be attenuated by the building, so the power density becomes:

$$S = \frac{F(\theta)}{4\pi} \frac{EIRP}{x^2 + h'^2}$$

For example, if the antenna is a half-wave dipole, the relative numeric gain is of the form of:

$$F(\theta, \phi) = \left[\frac{\cos\left(\frac{\pi}{2} \sin \theta\right)}{\cos \theta} \right]^2$$

Then, for a source with $EIRP$ of 1 000 W, the exposure power as a function of x is shown in Figure II.4 for three different relative heights $Dh = (h - h_2)$.

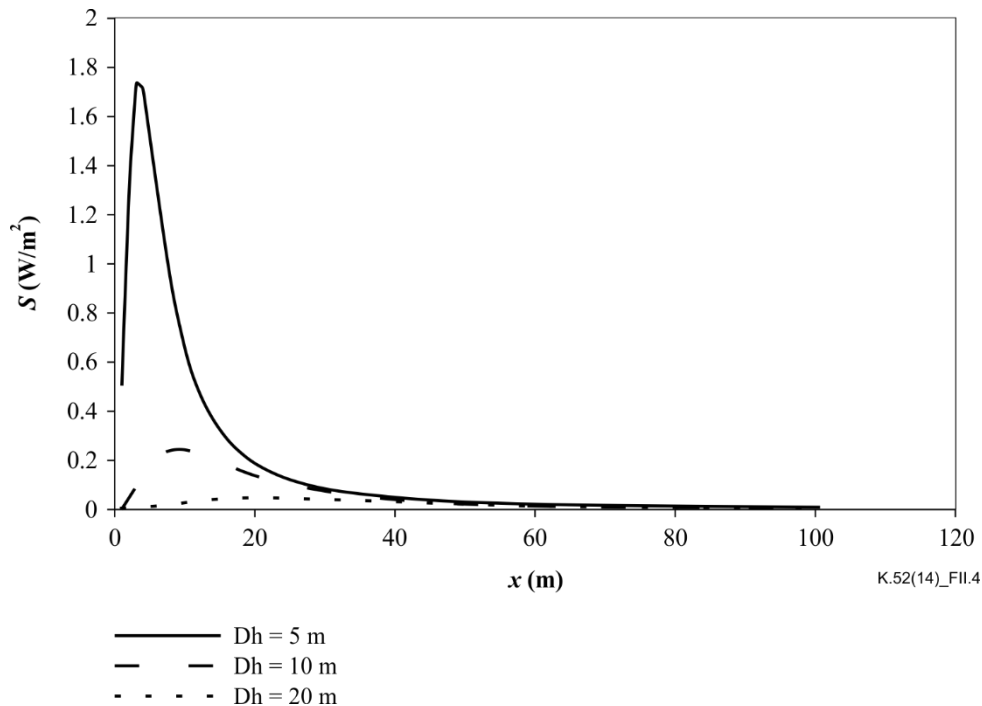


Figure II.4 – Power density at the ground level vs distance from the tower calculated for the example in Figure II.3

Appendix III

Example of $EIRP_{th}$ calculation

(This appendix does not form an integral part of this Recommendation.)

III.1 The $EIRP_{th}$ values

Tables III.1 to III.3 show the expressions for $EIRP_{th}$ values based on the [b-ICNIRP] limits for various frequency ranges, accessibility conditions and antenna directivity categories.

It is necessary to point out that the radiated density power can be used only in far-field conditions, when it is representative of the electric and magnetic fields. This represents the limit of validity of the proposed assessment procedure for normally compliant installations. Where the procedure is not applicable (e.g., low frequencies or exposure in near-field conditions), then the installation shall be considered provisionally compliant.

The ICNIRP guidelines define three frequency ranges to which correspond different limit values of equivalent plane wave power density. For frequencies above 100 MHz the limits are:

f (MHz)	$S_{lim}(f)$ (W/m ²)	
	General public	Occupational
100-400	2	10
400-2 000	$f/200$	$f/40$
$2 \cdot 10^3$ - $300 \cdot 10^3$	10	50

The $EIRP_{th}$ values are given as functions of antenna height and other relevant parameters (accessibility, directivity and frequency) defined in Annex B.

Appendix IV provides a rationale for the $EIRP_{th}$ values.

NOTE – In the following Tables a, d, h and h' are in metres.

Table III.1 – Conditions for normal compliance of installations based on [b-ICNIRP] limits for frequency range 100-400 MHz

Directivity category	Accessibility category	$EIRP_{th}$ (W)	
		General public	Occupational
1	1	$8\pi(h-2)^2$	$40\pi(h-2)^2$
	2	Lesser of: $8\pi(h-2)^2$ or $2\pi d^2$	Lesser of: $40\pi(h-2)^2$ or $10\pi d^2$
	3	Lesser of: $8\pi(h-2)^2$ or $2\pi \left[\frac{d^2 + (h-h')^2}{d} \right]^2$	Lesser of: $40\pi(h-2)^2$ or $10\pi \left[\frac{d^2 + (h-h')^2}{d} \right]^2$

Table III.1 – Conditions for normal compliance of installations based on [b-ICNIRP] limits for frequency range 100-400 MHz

Directivity category	Accessibility category	<i>EIRP_{th}</i> (W)	
		General public	Occupational
1	4	Lesser of: $8\pi(h-2)^2 \{ \text{If } a < (h-2) \}$ or $2\pi \left[\frac{a^2 + (h-2)^2}{a} \right]^2$	Lesser of: $40\pi(h-2)^2 \{ \text{If } a < (h-2) \}$ or $10\pi \left[\frac{a^2 + (h-2)^2}{a} \right]^2$
2	1	Lesser of: $\frac{2\pi}{A_{sl}}(h-2)^2$ or $2\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $10\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$
	2 (determined by: $h' > h - d \tan(\alpha + 1.129\theta_{bw})$)	Lesser of: $\frac{2\pi}{A_{sl}}(h-2)^2$ or $2\pi d^2$	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $10\pi d^2$
	3 (determined by: $h' < h - d \tan(\alpha + 1.129\theta_{bw})$)	Lesser of: $\frac{2\pi}{A_{sl}}(h-2)^2$ or $\frac{2\pi}{A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $\frac{10\pi}{A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$
	4	Lesser of: $\frac{2\pi}{A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $2\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{10\pi}{A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $10\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$
3	1	Lesser of: $\frac{2\pi}{A_{sl}}(h-2)^2$ or $2\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $10\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$

Table III.1 – Conditions for normal compliance of installations based on [b-ICNIRP] limits for frequency range 100-400 MHz

Directivity category	Accessibility category	<i>EIRP_{th}</i> (W)	
		General public	Occupational
3	2	N/A (Line of sight is usually required)	N/A (Line of sight is usually required)
	3 (determined by: $h' < h - d \tan(\alpha + 1.129\theta_{bw})$)	Lesser of: $\frac{2\pi}{A_{sl}}(h-2)^2$ or $\frac{\pi}{2A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $\frac{2.5\pi}{A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$
	4	Lesser of: $\frac{2\pi}{A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $2\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{10\pi}{A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $10\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$

Table III.2 – Conditions for normal compliance of installations based on [b-ICNIRP] limits for frequency range 400-2 000 MHz

Directivity category	Accessibility category	<i>EIRP_{th}</i> (W)	
		General public	Occupational
1	1	$\frac{f\pi}{50}(h-2)^2$	$\frac{f\pi}{10}(h-2)^2$
	2	Lesser of: $\frac{f\pi}{50}(h-2)^2$ or $\frac{f\pi}{200}d^2$	Lesser of: $\frac{f\pi}{10}(h-2)^2$ or $\frac{f\pi}{40}d^2$
	3	Lesser of: $\frac{f\pi}{50}(h-2)^2$ or $\frac{f\pi}{200} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$	Lesser of: $\frac{f\pi}{10}(h-2)^2$ or $\frac{f\pi}{40} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$

**Table III.2 – Conditions for normal compliance of installations based
on [b-ICNIRP] limits for frequency range 400-2 000 MHz**

Directivity category	Accessibility category	<i>EIRP_{th}</i> (W)	
		General public	Occupational
1	4	Lesser of: $\frac{f\pi}{50}(h-2)^2 \{ \text{If } a < (h-2) \}$ or $\frac{f\pi}{200} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$	Lesser of: $\frac{f\pi}{10}(h-2)^2 \{ \text{If } a < (h-2) \}$ or $\frac{f\pi}{40} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$
2	1	Lesser of: $\frac{f\pi}{200A_{sl}}(h-2)^2$ or $\frac{f\pi}{200} \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{f\pi}{40A_{sl}}(h-2)^2$ or $\frac{f\pi}{40} \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$
	2 (determined by: $h' > h - d \tan(\alpha + 1.129\theta_{bw})$)	Lesser of: $\frac{f\pi}{200A_{sl}}(h-2)^2$ or $\frac{f\pi}{200}d^2$	Lesser of: $\frac{f\pi}{40A_{sl}}(h-2)^2$ or $\frac{f\pi}{40}d^2$
	3 (determined by: $h' < h - d \tan(\alpha + 1.129\theta_{bw})$)	Lesser of: $\frac{f\pi}{200A_{sl}}(h-2)^2$ or $\frac{f\pi}{200A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$	Lesser of: $\frac{f\pi}{40A_{sl}}(h-2)^2$ or $\frac{f\pi}{40A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$
	4	Lesser of: $\frac{f\pi}{200A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $\frac{f\pi}{200} \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{f\pi}{40A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $\frac{f\pi}{40} \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$
3	1	Lesser of: $\frac{f\pi}{200A_{sl}}(h-2)^2$ or $\frac{f\pi}{200} \left[\frac{h}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{f\pi}{40A_{sl}}(h-2)^2$ or $\frac{f\pi}{40} \left[\frac{h}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$

Table III.2 – Conditions for normal compliance of installations based on [b-ICNIRP] limits for frequency range 400-2 000 MHz

Directivity category	Accessibility category	<i>EIRP_{th}</i> (W)	
		General public	Occupational
3	2	N/A (Line of sight is usually required)	N/A (Line of sight is usually required)
	3 (determined by: $h' < h - d \tan(\alpha + 1.129\theta_{bw})$)	Lesser of: $\frac{f\pi}{200A_{sl}}(h-2)^2$ or $\frac{f\pi}{50A_{sl}}\left[\frac{d^2 + (h-h')^2}{d}\right]^2$	Lesser of: $\frac{f\pi}{40A_{sl}}(h-2)^2$ or $\frac{f\pi}{10A_{sl}}\left[\frac{d^2 + (h-h')^2}{d}\right]^2$
	4	Lesser of: $\frac{f\pi}{200A_{sl}}\left[\frac{a^2 + (h-2)^2}{a}\right]^2$ or $\frac{f\pi}{200}\left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})}\right]^2$	Lesser of: $\frac{f\pi}{40A_{sl}}\left[\frac{a^2 + (h-2)^2}{a}\right]^2$ or $\frac{f\pi}{40}\left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})}\right]^2$

Table III.3 – Conditions for normal compliance of installations based on [b-ICNIRP] limits for frequency range 2 000-300 000 MHz

Directivity category	Accessibility category	<i>EIRP_{th}</i> (W)	
		General public	Occupational
1	1	$40\pi(h-2)^2$	$200\pi(h-2)^2$
	2	Lesser of: $40\pi(h-2)^2$ or $10\pi d^2$	Lesser of: $200\pi(h-2)^2$ or $50\pi d^2$
	3	Lesser of: $40\pi(h-2)^2$ or $10\pi\left[\frac{d^2 + (h-h')^2}{d}\right]^2$	Lesser of: $200\pi(h-2)^2$ or $50\pi\left[\frac{d^2 + (h-h')^2}{d}\right]^2$

**Table III.3 – Conditions for normal compliance of installations based on [b-ICNIRP] limits
for frequency range 2 000-300 000 MHz**

Directivity category	Accessibility category	<i>EIRP_{th}</i> (W)	
		General public	Occupational
1	4	Lesser of: $40\pi(h-2)^2 \{ \text{If } a < (h-2) \}$ or $10\pi \left[\frac{a^2 + (h-2)^2}{a} \right]^2$	Lesser of: $200\pi(h-2)^2 \{ \text{If } a < (h-2) \}$ or $50\pi \left[\frac{a^2 + (h-2)^2}{a} \right]^2$
2	1	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $10\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{50\pi}{A_{sl}}(h-2)^2$ or $50\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$
	2 (determined by: $h' > h - d \tan(\alpha + 1.129\theta_{bw})$)	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $10\pi d^2$	Lesser of: $\frac{50\pi}{A_{sl}}(h-2)^2$ or $50\pi d^2$
	3 (determined by: $h' < h - d \tan(\alpha + 1.129\theta_{bw})$)	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $\frac{10\pi}{A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$	Lesser of: $\frac{50\pi}{A_{sl}}(h-2)^2$ or $\frac{50\pi}{A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$
	4	Lesser of: $\frac{10\pi}{A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $10\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{50\pi}{A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $50\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$
3	1	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $10\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{50\pi}{A_{sl}}(h-2)^2$ or $50\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$

Table III.3 – Conditions for normal compliance of installations based on [b-ICNIRP] limits for frequency range 2 000-300 000 MHz

Directivity category	Accessibility category	<i>EIRP_{th}</i> (W)	
		General public	Occupational
3	2	N/A (Line of sight is usually required)	N/A (Line of sight is usually required)
	3 (determined by: $h' < h - d \tan(\alpha + 1.129\theta_{bw})$)	Lesser of: $\frac{10\pi}{A_{sl}}(h-2)^2$ or $\frac{2.5\pi}{A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$	Lesser of: $\frac{50\pi}{A_{sl}}(h-2)^2$ or $\frac{12.5\pi}{A_{sl}} \left[\frac{d^2 + (h-h')^2}{d} \right]^2$
	4	Lesser of: $\frac{10\pi}{A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $10\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$	Lesser of: $\frac{50\pi}{A_{sl}} \left[\frac{a^2 + (h-2)^2}{a} \right]^2$ or $50\pi \left[\frac{h-2}{\sin(\alpha + 1.129\theta_{bw})} \right]^2$
NOTE 1 – f is in MHz. NOTE 2 – All angles should be expressed in radians.			

A software calculator is provided as an electronic attachment to this Recommendation.

Appendix IV

Rationale for the $EIRP_{th}$ values of Tables in Appendix III

(This appendix does not form an integral part of this Recommendation.)

This appendix presents the rationale for $EIRP_{th}$ values in Appendix III. The rationale is based on calculations using far-field expressions for all cases. Therefore, the frequency range where this rationale applies is restricted to above 100 MHz.

IV.1 Inherently compliant sources

The criterion for the inherently compliant source is $EIRP$ of 2 W or less except for low-gain small aperture microwave or millimetre-wave antennas where total radiating power of 100 mW or less can be regarded as inherently compliant. This $EIRP$ corresponds to a power density of 0.16 W/m² at a distance of 1 m, while the lowest [b-ICNIRP] power density limit for the general public is 2 W/m².

IV.2 Normally compliant

The criteria for the normally compliant installations are derived by considering the exposure at ground level and at adjacent buildings or structures. A basic procedure for performing the calculation was shown in clause 10.1.2. The two determining factors are the antenna pattern and the accessibility conditions. For the derivation of the classification criteria, the following additional conservative assumptions are made:

- For ground level exposure, a reflection coefficient of 1 is assumed.
- All exposure is assumed to occur along the maximum of the antenna pattern in the horizontal plane.

The following subclauses show the derivation for the different antenna directivity categories.

IV.2.1 Directivity category 1

The antenna gain function is approximated by the relative numeric gain of an infinitesimal dipole.

$$F(\theta, \phi) = \left[\frac{\cos\left(\frac{\pi}{2} \sin \theta\right)}{\cos \theta} \right]^2 \approx \cos^2 \theta$$

The infinitesimal dipole has the broadest vertical gain function for an omnidirectional source. Thus, this represents the most severe exposure condition at the ground level with the main beam axis parallel to ground or higher.

Using this gain, the exposure power can be obtained analytically as a function of x , as:

$$S(x) = \frac{EIRP}{4\pi} \left(\frac{x}{x^2 + h_d^2} + \frac{x}{x^2 + h_s^2} \right)^2$$

where h_d is the difference between the height of the phase centre of the antenna, h , and the height of the observation point, and h_s is the sum of the quantities. The height of the observation point is 2 m for ground-level exposure and h' for exposure at adjacent structures. The calculation of maximum exposure is complicated, but a conservative estimate can be obtained by letting $h_s = h_d$. This approximation should be reasonably accurate near the surface, but produces a significant overestimate at points significantly above the surface. With this approximation, the maximum exposure occurs at $x = h_d$ and is equal to:

$$S_{max}(h) = \frac{1}{4\pi} \frac{EIRP}{h_d^2}$$

For a given limit, values of the equivalent plane wave power density, S_{lim} and a given antenna height, it is possible to calculate the maximum value of $EIRP$ that should provide compliance as:

$$EIRP_{th} = 4\pi h_d^2 S_{lim}$$

IV.2.2 Directivity category 2

In this case, the putative antenna pattern consists of two components; the main beam and the constant amplitude side-lobe envelope. The antenna pattern can be expressed as:

$$F(\theta) = \begin{cases} \left[\frac{\sin[c \sin(\theta - \alpha)]}{c \sin(\theta - \alpha)} \right]^2 & \text{main beam} \\ A_{sl} & \text{side - lobe envelope} \end{cases}$$

The parameter c determines the half-power beamwidth as follows:

$$c = \frac{1.392}{\sin(\theta_{bw}/2)}$$

The crossover from the main beam to the side-lobe region is difficult to evaluate analytically, however, it may be approximated as the first null of the main beam function. The first nulls occur at:

$$\theta_{n1,n2} = \alpha \pm \sin^{-1} \left[\frac{\pi}{1.392} \sin \left(\frac{\theta_{bw}}{2} \right) \right] \approx \alpha \pm 2.257 \frac{\theta_{bw}}{2}$$

Outside of the main beam, the exposure is calculated using the constant envelope, so that the maximum exposure occurs directly underneath the antenna. In many cases, this is a conservative assumption as the antenna pattern may have a null at this point. However, without additional pattern information, the most conservative assumption is being used. In some cases, the constant envelope can be modulated by a dipole factor ($\cos\theta$) for instance when a side-lobe exposure occurs far from the antenna base.

In addition, to simplify calculations, constant power in the main beam is assumed ($F(\theta) = 1$). The condition for a point (x,y) to be within the beam becomes:

$$h - x \tan \theta_{n1} \leq y \leq h - x \tan \theta_{n2}$$

IV.2.3 Directivity category 3

The main difference between exposure calculations for directivity category 3 compared to directivity category 2 concerns the treatment of the reflected field. Antennas in directivity category 3 are used for point-to-point links, so that it is not necessary to consider reflected waves for exposure in the main beam.

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